

Summary

Assignment is to design a revolute compliant joint and a translational complaint joint. This file shows four attempts and two designs of revolute one.

Attempt 1: Origami

Origami can be manufactured out of sheet material by creasing and folding. I first designed following 1DOF mechanism. (Figure 1)

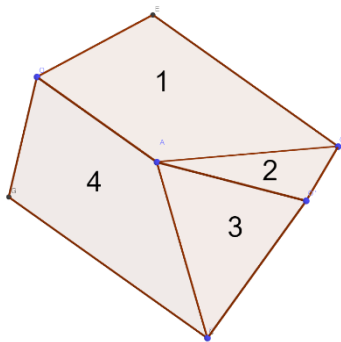


Figure 1 Link 1 and link 4 rotates about the intersection, thus forming a revolute joint. For animation, please see <https://www.geogebra.org/m/ca9bangx>

The designed rotation axis is the intersection between 1 and 4. The mechanism has two parallel chains: 1234 and 14. Unfortunately, chain 1234 does not increase out-of-axis stiffness. Only chain 14 provide out-of-axis stiffness. Thus, the useful part is basically one lumped joint between 1 and 4, and thus is not a good candidate.

Attempt 2: Symbolic Synthesis

Freudenstein and Maki's work discussed a mechanism synthesis method with graph theory symbol, and I tried applying them. However, symbolic synthesis disregard symmetry, while symmetry is very common in mechanisms.

For example, for planar double parallelogram mechanisms, symbolic synthesis gives wrong degree of freedom (Figure 1 and [1])

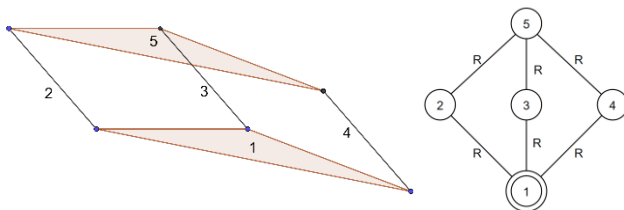


Figure 2 Left: bar represents link connected by rotation joint; Right: Symbolic notation of left mechanism.

$$F = \lambda(l - j - 1) + \sum_{i=1}^j f_i$$

$$= 3 \times (5 - 6 - 1) + 6 \times 1$$

$$= 0 \quad [1]$$

The situation worsen for spatial mechanism. Take previous origami as example and take four plate in Figure 1 as four link (Figure 3).

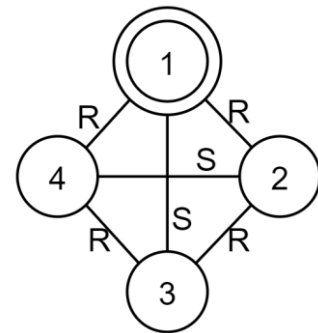


Figure 3 each circle represent a plate with corresponding number in Figure 1

Equation gives wrong degree of freedom [2].

$$F = \lambda(l - j - 1) + \sum_{i=1}^j f_i$$

$$= 6 \times (4 - 6 - 1) + 4 \times 1 + 2 \times 3$$

$$= -8 \quad [2]$$

I tried again, decomposing plate into bars to cancel symmetry. I was expecting the decomposition to yield a positive degree of freedom, indicating an under-constraint mechanism (Figure 4).

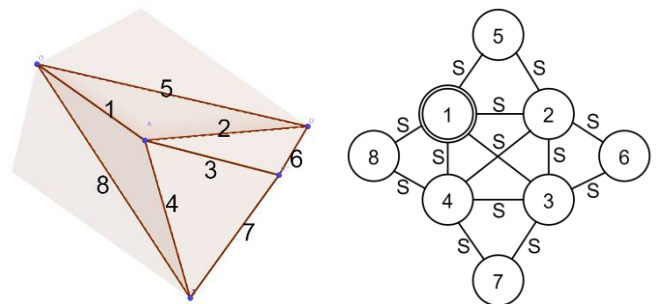


Figure 4 All plates are decomposed into 3 bar connected interconnected with 3 ball joints. This introduce local degree of freedom in longitudinal rotation of bars.

However, equation gives unwanted result [3]

$$F = 6 \times (8 - 14 - 1) + 14 \times 3 = 0 \quad [3]$$

To sum up, I failed to get the symbolic synthesis work on spatial mechanism.

Attempt 3: Ring

The first time I 3D-printed a ring (Figure 5), I thought it can be redesigned into a revolute joint.

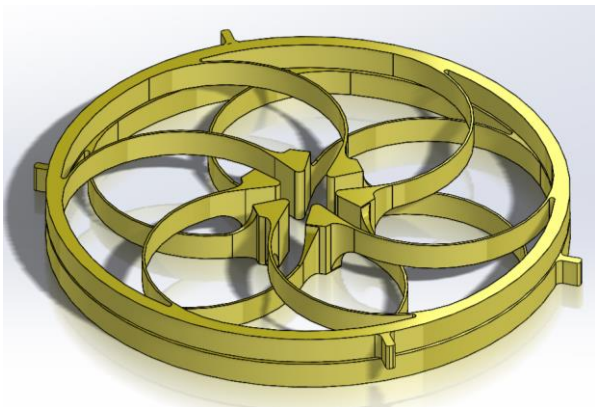


Figure 5 The ring I CAD and hanged on the handle of your office last year

Beam has large tensile stiffness but small shearing stiffness. I think Trease's design exploited that feature, thus creating large stiffness ratio for out of motion-axis loading condition (Figure 6).

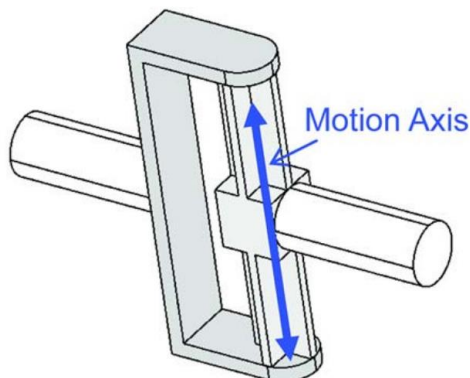


Figure 6 Under and only under torque about motion axis, all beams are under shearing load and no tensile load. Under all load out of motion-axis, at least one beam is under tension.

Trying to exploit same feature, I re-CAD the ring (Figure 7). It has around 30mm diameter and 3 stages. Upper and lower stages are fixed. Middle stage rotates.

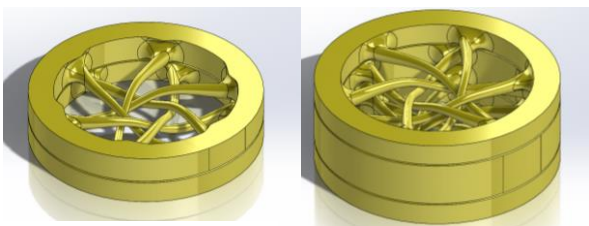


Figure 7 The left is half of the revolute joint. Mirroring it about upper surface, then we got the whole revolute joint on the right. Beams are arranged to take tensile load under out of motion-axis displacement

ABS is temporarily chosen as material to run FEA. The ring is CADed with dimensionless design

parameter. Thus, I can easily re-scale the ring or change shape of inner beam for different materials

However, FEA bring bad news. Under only 10N of axis load, the joint has 0.4mm axial displacement (Figure 8).

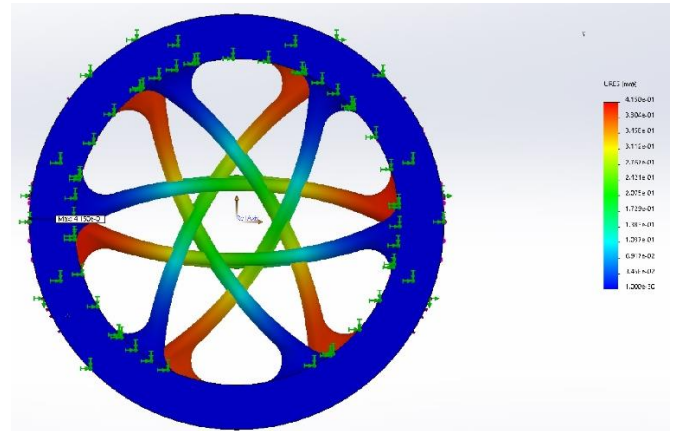


Figure 8 Linear static finite element analysis with SolidWorks. Upper and lower stage fixed, on middle stage applied 10N axial load, 0.4mm axial displacement is found on middle ring.

Attempt 4: Ring Again

I rearrange the beam, decreasing the length and increasing the slope (Figure 9).

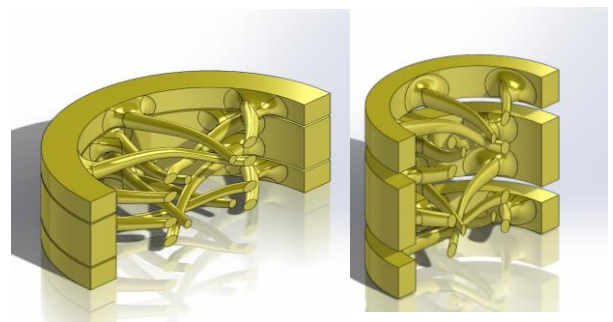


Figure 9 Comparison of inner structure between and after beam rearrangement.

Run FEA again and again, the result is as follows (Table 1).

Table 1 FEA results. Torsion torque and axial force are around motion axis. Bending torque and is radius force are perpendicular to motion axis. In lower table, $k_{rot} = 2k/d$, d is the diameter of joint.

	Load M [Nm]	Translation d [mm]	Rotation φ [rad]	Rot. Stiffness k_{rot} [Nm/rad]
Torsion	0.8	4.602	0.778	1.02
Bending	0.8	0.473	0.080	9.98

	Load M [N]	Translation d [mm]	Stiffness k [N/m]	Rot. Stiffness k_{rot} [Nm/rad]
Axial	20.	0.083	2.40E+05	53.99
Radius	20.	0.076	2.63E+05	59.27

Thus, we get following comparison with Trease's result (Table 2).

Table 2 Comparison between ring design and Trease's design in coefficient between stiffness and torsion stiffness

	Ring Again FEA	Trease FEA
Torsion Co.	1.	1
Bending Co.	9.78	292
Axial Co.	52.93	1194
Radius Co.	58.11	38017

The comparison is frustrating. What is even more frustrating is that I later realize the ring is similar to Trease's design in topology. If the ring has 4 beams on each sides instead of 6, then the ring has exactly same topology with that of Trease.

No matter how I redesign the ring, it's still trapped in the topology of Trease's design. I now come to appreciate Trease's design.